



### 3D-Printed Robots That Walk Without Electronics

Researchers at UC San Diego's Bioinspired Robotics Lab have developed a six-legged robot that walks without electronics, powered solely by compressed gas. Fully 3D-printed in one step using a single flexible material, the robot



*This robot can walk, without electronics, and only with the addition of a cartridge of compressed gas, right off the 3D-printer (Credit: David Baillot/ University of California San Diego)*

includes integrated soft actuators and a pneumatic oscillating circuit that mimics a steam engine to control motion. Tested indoors and outdoors, it operated untethered across turf, sand, and underwater, and ran continuously for three days with a steady gas supply. Future designs aim to add onboard gas storage, biodegradable materials, and functional appendages like grippers.

### Nuclear Batteries That Last For Decades

Researchers from Daegu Gyeongbuk Institute of Science & Technology (South Korea) are developing nuclear batteries using radiocarbon as a safer, long-lasting alternative to lithium-ion batteries. Led by Su-Il In at DGIST, the team created a prototype betavoltaic battery using carbon-14 and a titanium dioxide-based semiconductor sensitised with a ruthenium dye. By placing radiocarbon on both the cathode and anode, energy-conversion efficiency increased from 0.48% to 2.86%. These batteries could power devices for decades, reducing environmental impact and minimising recharging needs.

### Neuromorphic Computing With Standard Silicon Transistor

Researchers at the National University of Singapore have shown that a single standard silicon transistor can mimic biological neurons and synapses. This marks a significant step in neuromorphic computing by enabling in-

memory computing using commercial CMOS technology. Their device, Neuro-Synaptic RAM (NS-RAM), switches between neuron-like firing and synaptic weight changes via punch-through impact ionisation and charge trapping. NS-RAM offers low power use, stability, and scalability, promising more efficient, brain-like AI hardware for future processors.

### Slow Current Limits Organic Solar Efficiency

Researchers at Chemnitz University of Technology (Chemnitz, Germany) have enhanced organic solar cell efficiency using novel semiconductors and printable 'ink' technology. Unlike crystalline silicon, these lightweight, low-cost cells suffer from slow charge transport due to their disordered active layers. This increases transport resistance, reducing fill factor and power output. By analysing current-voltage behaviour and comparing it with the suns-Voc method, the team identified transport resistance as a significant but surmountable limitation. Further studies reveal that the shape of the density of states also impacts efficiency significantly.

### Enhancing MIS With Tactile Sensing Technology

Minimally invasive surgery (MIS) offers faster recovery and lower infection risk but lacks tactile feedback, making precision difficult. NYU Abu Dhabi researchers have developed an 'off-the-jaw' sensor design that restores touch



*Enhancing MIS with tactile sensing technology (Credit: NYU Abu Dhabi)*

without placing sensors near surgical tool tips. This reduces contamination, simplifies sterilisation, and integrates with standard laparoscopic instruments. The device provides real-time tissue stiffness and thickness

data, improving safety and precision, especially for trainees. Early trials show a 30% efficiency boost. Plans include adapting the technology for robotic-assisted surgery and enhancing sensors for better tissue differentiation.